



DRISHTI-AI: Diabetic Retinopathy Intelligent Screening with Hierarchical Triage & Interpretability

An Explainable Edge-AI Retinal Screening and Clinical Decision Support System for Rural Primary Health Centres

Problem Statement ID: SIH26038 • **Domain:** Healthcare, Telemedicine & Rural Public Health • **Standard:** ABDM / FHIR R4 & NPCB&VI

Team Size

6 Members

3 ML • 2 DS • 1 Full Stack

Core Platform

MATLAB R2021b+

App Designer & Deep Learning

Target Standard

ABDM / FHIR R4

NPCB&VI SaMD Class-B

Target Latency

< 50 ms

Standard Edge CPU

● 1. PROBLEM STATEMENT & CLINICAL CONTEXT

Diabetic Retinopathy (DR) is a microvascular complication of diabetes mellitus that damages the retinal capillaries, leading to progressive and irreversible visual impairment if untreated. India is currently recognized as the "diabetes capital of the world," with an estimated **77 to 101 million individuals** diagnosed with diabetes and over **136 million** in pre-diabetic stages. Epidemiological studies indicate that approximately **18% to 22% of diabetic patients in India develop diabetic retinopathy**, with 5% to 7% progressing to sight-threatening stages.

Key Rural Healthcare Bottlenecks:

- **Specialist Deficit:** Over 80% of India's ophthalmologists practice in tier-1/tier-2 metropolitan centres, whereas ~70% of the diabetic population resides in rural villages and remote districts.
- **Equipment & Operator Inexperience:** Primary Health Centres (PHCs) lack trained optometrists. Community health workers (ASHAs/ANMs) using portable cameras frequently capture blurred, underexposed, or off-center fundus images, risking severe false negatives.
- **Zero Connectivity:** Cloud-based deep learning APIs are unviable in rural health camps due to absent or intermittent cellular coverage.
- **The "Black Box" Barrier:** Conventional neural networks output raw probability scores without visual explanations, preventing local doctors from validating why an AI flagged a retina as diseased.

Formulated Project Objective (SIH26038):

To design, implement, and validate an edge-deployable, offline-first, explainable artificial intelligence (XAI) screening platform capable of autonomously grading fundus photographs into 5 ICDR stages, verifying capture quality via mathematical image quality metrics, visually highlighting lesion locations using Grad-CAM, and generating automated clinical referral documentation compliant with National Programme for Control of Blindness & Visual Impairment (NPCB&VI) and Ayushman Bharat Digital Mission (ABDM) standards.

● 2. PROPOSED SOLUTION ARCHITECTURE

To address these challenges, we propose an integrated software engineering solution comprising **six modular subsystems** running completely locally on edge hardware (standard laptops) with zero cloud dependency:

// DRISHTI-AI Proposed Pipeline Flow

- [1. Ingestion & Data] → (APTOS-2019 / EyePACS / IDRiD multi-benchmark standardization)
- [2. Optical IQA Gate] → (Laplacian Blur, Illumination Uniformity, RMS Contrast, Immerkaer Noise)
- [3. Preprocessing] → (L*a*b* Rayleigh CLAHE + Morphological Illumination Flattening)
- [4. Transfer CNN Model] → (ResNet-50 / MobileNetV2 with Custom Retinal Head & Softmax)
- [5. Explainable AI] → (Grad-CAM Backpropagation + Otsu Lesion Bounding Boxes + Quadrants)
- [6. Reporting & ABDM] → (A4 Vector PDF with Quad-Panel + ABDM FHIR R4 JSON Export)
- [7. Workstation GUI] → (18-Screen MATLAB App Designer with Dark/Light Mode & Calipers)

● 3. TEAM ROSTER & PLANNED WORK BREAKDOWN (6 MEMBERS)

To ensure modularity and accountability, the project responsibilities are divided across **3 Machine Learning students, 2 Data Science students, and 1 Full Stack student:**

S.No	Student Name	Registration No.	Domain	Primary Subsystems Owned
1	Subham Panigrahi	12312794	Data Science	Dataset Ingestion, Class Balance & 5-Metric IQA Gatekeeper
2	Konduri Mrunal	12316339	Data Science	L*a*b* CLAHE Retinal Enhancement & Simulink Queue Simulation
3	Rajbardhan Kumar	12326119	Machine Learning	Transfer Learning CNN Backbones, Head Surgery & Training Loop
4	Priyam Saxena	12313674	Machine Learning	Real-Time Edge Inference Engine, Clinical Triage & Validation
5	Kadambala Likhith	12314034	Machine Learning	Grad-CAM Saliency Engine, Lesion Bounding & Text Justifications
6	Vaibhav Raj	12325142	Full Stack	18-Screen Workstation GUI, A4 PDF Reports & ABDM FHIR R4 Hub

Subham Panigrahi

Reg. No: 12312794

Role: Dataset Engineering & 5-Metric IQA Gate

Modules: data/ & qualityAssessment/

- Standardize multi-source benchmarks (APTOS, EyePACS, IDRiD, Messidor).
- Implement stratified 70/15/15 train/val/test splits for clinical class balance.
- Build synthetic fundus generator for offline test suite automation.
- Formulate 5 mathematical IQA metrics (Laplacian blur, illumination balance, RMS contrast, Immerkaer noise, Tenengrad sharpness).

Data Science

Konduri Mrunal

Reg. No: 12316339

Role: Retinal Enhancement & Queue Simulation

Modules: preprocessing/ & simulink/

- Build 6-stage retinal enhancement pipeline in L*a*b* (Lightness-Color) color space.
- Implement Rayleigh CLAHE and morphological background flattening.
- Calculate objective image metrics (target: PSNR > 31 dB, SSIM > 0.94).
- Develop Simulink discrete-event queue models (single-camera, multi-camera, and realistic queueing) to simulate rural camp throughput.

Data Science

Rajbardhan Kumar

Machine Learning

Priyam Saxena

Machine Learning

Reg. No: 12326119

Role: Deep Transfer Learning & Training

Modules: classification/buildModel.m & training/

- Configure transfer learning backbones (ResNet-50, MobileNetV2, EfficientNet-B0).
- Perform network surgery: GAP + Dropout (40% deactivation rate) + 5-unit Softmax head.
- Configure stochastic data augmentation (affine rotations, scaling, shear).
- Implement training loop with Adam, piecewise LR decay, and early stopping.

Reg. No: 12313674

Role: Real-Time Edge Inference & Validation

Modules: classification/predictDR.m & testing/

- Build high-speed edge inference engine targeting < 50 ms latency on CPU.
- Program clinical referral triage logic (Stages 0–1 routine vs Stages 2–4 referral).
- Validate multi-class metrics: 5x5 confusion matrix, One-vs-Rest ROC/AUC.
- Evaluate Cohen's Quadratic Weighted Kappa (target score ≥ 0.90) and clinical sensitivity.

Kadambala Likhith

Reg. No: 12314034

Machine Learning

Role: Explainable AI (Grad-CAM) & Lesions

Modules: explainability/

- Implement Grad-CAM backpropagation on final convolutional feature maps.
- Overlay alpha-blended Jet/Turbo heatmaps on retinal fundus captures.
- Apply Otsu thresholding and connected component analysis to segment lesions.
- Automate quadrant localization and plain-language clinical justification generation.

Vaibhav Raj

Reg. No: 12325142

Full Stack

Role: Workstation GUI, Reports & Integration

Modules: gui/, reports/, main.m, & utils/

- Build 18-screen clinical workstation in MATLAB App Designer (Dark/Light theme).
- Implement canvas measurement calipers for micro-lesion inspection in micrometers (μm).
- Engineer NPCB&VI-compliant A4 clinical PDF reports with Quad-Image diagnostic panel.
- Build Ayushman Bharat (ABDM) FHIR R4 JSON export hub and orchestrator ('main.m').

● 4. PLANNED TECHNICAL APPROACH & METHODOLOGY

4.1 Optical Image Quality Assessment (IQA Gatekeeper)

Pre-Screening Quality Filter

Problem Solved: Community health workers in rural camps often capture blurred, out-of-focus, or poorly lit retinal pictures. Feeding sub-standard images directly into AI causes catastrophic false-negative diagnoses.

1. Blur & Focus: Evaluates edge sharpness via modified Laplacian variance to catch patient movement or blinking.

2. Illumination Balance: Measures brightness across 4 quadrants to detect uneven flash or peripheral shadowing.

3. Vessel Contrast: Root Mean Square (RMS) contrast checks if micro-capillaries are clearly distinguishable.

4. Noise & Sharpness: Immerkaer noise estimation and Tenengrad gradient energy accumulation.

Outcome: The system automatically outputs a 3-way verdict: **Good (Proceed)**, **Needs Enhancement**, or **Retake Image** (alerting the operator immediately while the patient is still seated).

4.2 Retinal Feature Enhancement in L*a*b* Color Space

Contrast Optimization

Problem Solved: Early microaneurysms and faint capillary hemorrhages have very low contrast against the reddish retina. Enhancing contrast directly in standard RGB distorts color balance, turning blood vessels unnatural shades of purple or cyan.

What is the L*a*b* Color Space? (Decoupling Brightness from Color)

Unlike RGB (where brightness and colors are mixed across all channels), **L*a*b*** separates an image into:

- **L* (Lightness / Luminance):** Measures pure brightness from 0 (pure black) to 100 (diffuse white).
- **a* (Green-Red Axis):** Encodes the green-to-red color spectrum of retinal tissue.
- **b* (Blue-Yellow Axis):** Encodes the blue-to-yellow color spectrum.

Proposed Enhancement: We convert the fundus capture to L*a*b* and apply **Contrast Limited Adaptive Histogram Equalization (CLAHE)** with Rayleigh clipping *strictly to the L* (Lightness) channel*. The a* and b* color channels are left untouched.

Clinical Result: Subtle micro-lesions become sharply visible with high contrast, while the retina retains its natural warm reddish-orange clinical hue without any color distortion.

4.3 Deep Transfer Learning & 5-Stage ICDR Staging

AI Severity Grading

Proposed Architecture: We implement transfer learning utilizing pre-trained convolutional backbones:

- **MobileNetV2 (3.5M parameters):** Inverted residual bottlenecks engineered for real-time edge CPU inference (< 50 ms) on standard laptops without requiring an expensive GPU.
- **ResNet-50 (25.6M parameters):** Deep residual architecture providing a high-capacity clinical validation benchmark.

Custom Retinal Head Surgery: The original ImageNet 1000-class layer is replaced with Global Average Pooling (GAP), a Dropout layer ($p = 0.40$ to prevent clinical overfitting), and a 5-neuron Softmax classification head outputting calibrated probabilities for:

Grade 0: Normal (No DR)

Grade 1: Mild NPDR

Grade 2: Moderate NPDR

Grade 3: Severe NPDR

Grade 4: Proliferative DR

4.4 Explainable AI (Grad-CAM) & Visual Lesion Localization

Interpretability Engine

Problem Solved: Neural networks are notoriously "black boxes". A medical practitioner cannot trust an AI that outputs a raw grade without showing *where* in the patient's eye it identified pathology.

How Grad-CAM Works in 2 Intuitive Steps:

Step 1: Compute Feature Importance Weights (α)

The algorithm traces gradients from the predicted disease score back to the final convolutional layer. It evaluates: "*How strongly did each feature channel contribute to flagging this retina as diseased?*" Gradients are average-pooled across all pixels to produce a single importance score for each feature channel.

Step 2: Generate the Visual Heatmap ($L_{\text{Grad-CAM}}$)

The feature channels are multiplied by their importance weights and summed together. A **ReLU** (Rectified Linear Unit) filter is applied to discard negative values, keeping *only the positive visual evidence* of lesions (such as microaneurysms, hemorrhages, and hard exudates).

Clinical Output: The 2D heatmap is overlaid onto the fundus photograph (Red/Yellow = disease hotspots, Blue = healthy tissue), automated bounding boxes are drawn around clustered lesions, and plain-language notes indicate the affected quadrant (e.g., "*Microaneurysms localized in Inferotemporal region*").

● 5. TECHNOLOGY STACK & PLANNED TOOLING

Domain	Proposed Tool / Library	Technical Rationale
Core Environment	MATLAB R2021b+	Native medical device engineering, signal processing, and FDA validation support.
Deep Learning	Deep Learning Toolbox	LayerGraph manipulation, transfer learning, autograd for Grad-CAM, and CPU optimization.
Image Processing	Image Processing Toolbox	L*a*b* conversion, adaptive histogram equalization (CLAHE), morphological filtering.
Desktop GUI	MATLAB App Designer	18-view responsive clinical workstation, theme switching, and interactive calipers.
Operations Research	Simulink & SimEvents	Discrete-event queueing simulation modeling patient arrival and camera bottlenecks.
Healthcare Standards	HL7 FHIR R4 & NPCB&VI	Standardized clinical PDF reports and JSON export for Ayushman Bharat (ABDM) integration.

● 6. EXPECTED DELIVERABLES & IMPACT

1. Fully Integrated MATLAB System Executable via an interactive 18-screen App Designer workstation and automated headless CLI (<code>main.m</code>).	2. Clinical Validation Benchmark Target Quadratic Weighted Kappa $\kappa_w \geq 0.90$, Sensitivity $\geq 90\%$, Specificity $\geq 90\%$, and edge latency < 50 ms.
3. Hospital-Grade Reporting Hub Automated A4 diagnostic PDF reports with Quad-Image panel and ABDM-compliant FHIR R4 JSON records.	4. Complete Engineering Suite Full documentation suite including Software Requirements Specification (SRS), System Design Document (SDD), and FMEA risk analysis.

Project: **DRISHTI-AI (SIH26038)** • 6-Member Capstone Engineering Team
Submission: Capstone Project Proposal & Implementation Plan

Academic Year: 2026
Document Reference: PROPOSAL-DRISHTI-SIH26038-REV2